Advanced Algebra

ANSWER KEY

Section 1: Polynomial Theorems & Synthetic Division

1 If $(x-2)$ is a factor of $P(x) = 2x^3 - 3x^2 + kx - 10$, what is the value of k?

a. 1

b. 2

c. 3 ✓

d. 4

By the Factor Theorem, $P(2) = 0$:
$2(8) - 3(4) + 2k - 10 = 0 \Rightarrow 16 - 12 + 2k - 10 = 0 \Rightarrow 2k - 6 = 0 \Rightarrow k = 3$.

2 What is the quotient and remainder when $3x^3 + 4x^2 - x + 5$ is divided by $(x+2)$?

a. **Quotient** $3x^2 - 2x + 3$**, remainder** -1 ✓

b. Quotient $3x^2 + 2x - 3$, remainder 1

c. Quotient $3x^2 - 2x - 3$, remainder 11

d. Quotient $3x^2 + 10x + 19$, remainder 43

Using synthetic division with -2: coefficients $[3, 4, -1, 5]$. Drop 3: $-2(3) = -6$, $4 - 6 = -2$; $-2(-2) = 4$, $-1 + 4 = 3$; $-2(3) = -6$, $5 - 6 = -1$. Quotient is $3x^2 - 2x + 3$, remainder -1.

3 Find the sum of the cubes of the roots of $x^2 - 4x + 2 = 0$.

a. 32

b. 40 ✓

c. 48

d. 52

By Vieta's: $r + s = 4$ and $rs = 2$. $r^3 + s^3 = (r+s)^3 - 3rs(r+s) = 4^3 - 3(2)(4) = 64 - 24 = 40$.

Section 2: Rational Equations & Partial Fractions

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4 Solve the rational equation: $\frac{x}{x-2} + \frac{2}{x+2} = \frac{8}{x^2-4}$.

a. $x = 2$

b. $x = -2$

c. $x = 2$ and $x = -2$

d. **No real solution ✓**

Multiply by $(x-2)(x+2)$: $x(x+2) + 2(x-2) = 8 \Rightarrow x^2 + 4x - 12 = 0 \Rightarrow (x+6)(x-2) = 0$. The roots are $x = -6$ and $x = 2$. However, $x = 2$ is an extraneous root (denominator zero).

5 Decompose $\frac{8x-4}{x^2-4}$ into partial fractions $\frac{A}{x-2} + \frac{B}{x+2}$. What is A?

a. 2

b. 3 ✓

c. 4

d. 5

$8x - 4 = A(x+2) + B(x-2)$. Let $x = 2$: $8(2) - 4 = A(4) \Rightarrow 12 = 4A \Rightarrow A = 3$.

6 Simplify the compound fraction: $\frac{\frac{1}{x+h} - \frac{1}{x}}{h}$ for $h \neq 0$.

a. $-\frac{1}{x(x+h)}$ ✓

b. $\frac{1}{x(x+h)}$

c. $-\frac{1}{x^2}$

d. $\frac{h}{x(x+h)}$

Combine fractions in numerator: $\frac{x-(x+h)}{x(x+h)h} = \frac{-h}{hx(x+h)} = -\frac{1}{x(x+h)}$.

Section 3: Radical Equations & Fractional Exponents

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7 Solve for x: $\sqrt{2x+9} - \sqrt{x+1} = 2$.

a. 0

b. 8

c. 0 **and** 8 ✓

d. No real solution

Rearrange: $\sqrt{2x+9} = \sqrt{x+1} + 2$. Squaring both sides:
$2x+9 = (x+1) + 4\sqrt{x+1} + 4 \Rightarrow x+4 = 4\sqrt{x+1}$. Squaring again:
$x^2 + 8x + 16 = 16x + 16 \Rightarrow x^2 - 8x = 0 \Rightarrow x(x-8) = 0$. Both $x = 0$ and $x = 8$ are valid.

8 If $x^{2/3} - 5x^{1/3} + 6 = 0$, what are all real values of x?

a. 8 **and** 27 ✓

b. 2 and 3

c. ± 8

d. ± 27

Let $u = x^{1/3}$: $u^2 - 5u + 6 = 0 \Rightarrow (u-2)(u-3) = 0 \Rightarrow u = 2$ or $u = 3$. Cubing gives $x = 2^3 = 8$ or $x = 3^3 = 27$.

9 Simplify the radical quotient: $\frac{x-8}{\sqrt[3]{x}-2}$ for $x \neq 8$.

a. $x^{2/3} + 2x^{1/3} + 4$ ✓

b. $x^{2/3} - 2x^{1/3} + 4$

c. $x^{2/3} + 4$

d. $(\sqrt[3]{x} + 2)^2$

Use difference of cubes factorization $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$ where $a = \sqrt[3]{x}$ and $b = 2$.

Section 4: Non-Linear Systems & Inequalities

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10 Solve the non-linear system: $y = x^2 - 2x - 3$ and $y = 2x + 2$. What is the sum of the x-coordinates of the intersection points?

a. 2

b. 4 ✓

c. -4

d. 6

Set equations equal: $x^2 - 2x - 3 = 2x + 2 \Rightarrow x^2 - 4x - 5 = 0$. By Vieta's formulas, the sum of roots is $-\frac{-4}{1} = 4$.

11 What is the solution set of the quadratic inequality $2x^2 - 5x - 3 > 0$?

a. $(-\infty, -\frac{1}{2}) \cup (3, \infty)$ ✓

b. $(-\frac{1}{2}, 3)$

c. $(-\infty, -3) \cup (\frac{1}{2}, \infty)$

d. $x > 3$

Factor: $(2x + 1)(x - 3) > 0$. Roots are $x = -\frac{1}{2}$ and $x = 3$. Testing intervals shows the inequality holds outside the roots: $(-\infty, -\frac{1}{2}) \cup (3, \infty)$.

12 For what values of m will the line $y = mx - 1$ NOT intersect the parabola $y = x^2$?

a. $(-2, 2)$ ✓

b. $[-2, 2]$

c. $(-\infty, -2) \cup (2, \infty)$

d. No such values of m

$x^2 = mx - 1 \Rightarrow x^2 - mx + 1 = 0$. No intersection requires discriminant $D < 0$:
$m^2 - 4(1)(1) < 0 \Rightarrow m^2 < 4 \Rightarrow -2 < m < 2$.

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